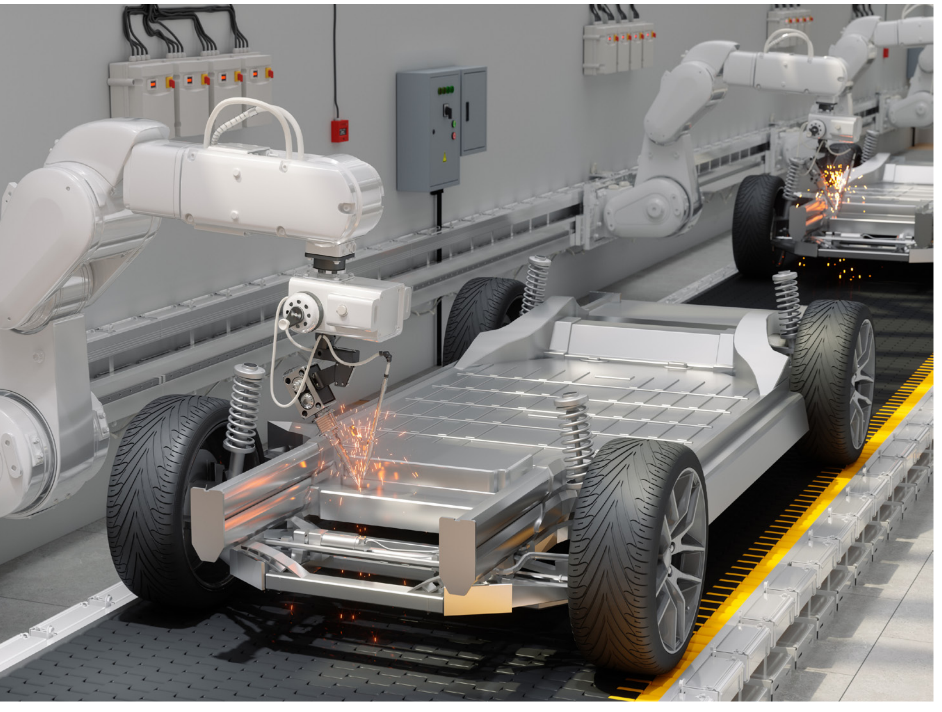
Lithium-metal batteries can double the EV range



Stanford scientists have unveiled a breakthrough in lithium metal battery technology that could pave the way for cars to travel 800 to 1130km (500 to 700 miles) on a single charge (Credit: www.istockphoto.com)

Researchers at Stanford University have developed lithium metal battery technology, potentially doubling the range of electric cars to 800 to 1130km (500 to 700 miles) on a single charge. Unlike lithium-ion batteries, which use graphite anodes and lithium metal oxide cathodes, lithium-metal batteries employ electroplated lithium metal for the anode, significantly boosting energy storage capacity. However, these batteries face a challenge: their energy storage capacity rapidly declines after a few charge-discharge cycles, primarily due to forming a solid electrolyte interphase (SEI) matrix that captures isolated lithium metal fragments. The team discovered a simple yet effective solution: discharging the battery and letting it rest for a few hours. This process allows the SEI matrix to dissolve, reviving inactive lithium fragments and enhancing battery performance without requiring expensive equipment or material changes.

Vision sensors with expanded field of view for robots and drones

Researchers at Beihang University in China have developed a multi-camera differential binocular vision sensor with an expanded field of view (FOV). This new sensor is designed to improve the accuracy of measurements for robots, drones, autonomous vehicles, and other smart systems. Traditional sensors often use single cameras, which can limit their measurement precision due to a restricted FOV. The innovative sensor consists of a central primary camera surrounded by four peripheral auxiliary cameras, creating four binocular pairs for more precise three-dimensional measurements. The sensor's structure parameters have been optimised for spatial arrangement, measurement range, and accuracy, resulting in a much wider FOV than conventional binocular cameras.

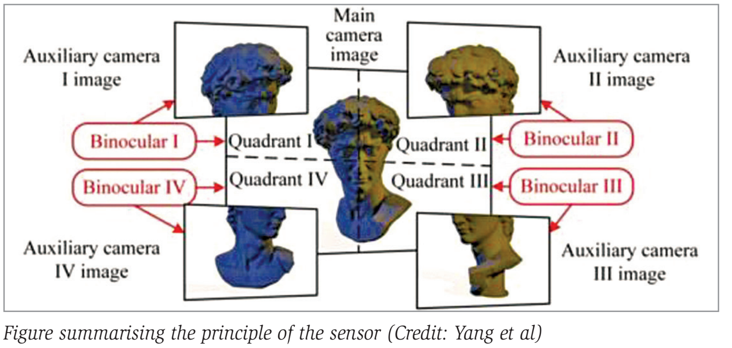


Figure summarising the principle of the sensor (Credit: Yang et al)